

# A Literature-Implied Positive Subcase of the Uniform Interpolating-Blaschke Approximation Problem

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**Status.** *Literature-implied answer (partial subcase).* The implication below is an agent-identified combination of known theorems. It is not presented as a new full solution.

## Source question

Chalendar, Gorkin, and Partington ask on PDF page 4 of [1]:

Can every Blaschke product be approximated (uniformly) by an interpolating Blaschke product?

Here “uniformly” means in the  $H^\infty$  norm on the unit disk. Xin and Hou still describe the corresponding density problem for the full inner-function space as open in their 2023 introduction [4]. The two formulations are equivalent at the level of density because Frostman’s theorem uniformly approximates every inner function by Blaschke products.

## The literature bridge

For  $a \in \mathbb{D}$ , put

$$\varphi_a(w) = \frac{a - w}{1 - \bar{a}w}.$$

Following Mortini and Nicolau, let  $\mathcal{M}$  be the class of inner functions  $\Theta$  such that  $\varphi_a \circ \Theta$  is a Carleson–Newman Blaschke product for every  $a \neq 0$ . Thus every such shift is a finite product of interpolating Blaschke products. Marshall and Stray prove that a finite product of interpolating Blaschke products can be approximated arbitrarily closely in  $H^\infty$  norm by a single interpolating Blaschke product; see the proof of their Lemma 1, especially journal pages 494–495 [2]. The modern paper [3] records the class  $\mathcal{M}$  in equation (2), proves that every SIP inner function lies in  $\mathcal{M}$  (Corollary 1.6(a)), and supplies useful examples and geometric subclasses.

**Theorem 1.** *Every inner function  $\Theta \in \mathcal{M}$  is a uniform limit in  $H^\infty$  of interpolating Blaschke products. More quantitatively, for every  $a \in \mathbb{D}$  with  $a \neq 0$ , the function*

$$F_a = \frac{\Theta - a}{1 - \bar{a}\Theta}$$

*is Carleson–Newman and*

$$\|F_a - \Theta\|_\infty \leq \frac{2|a|}{1 - |a|}. \tag{1}$$

*Proof.* Since  $F_a = -\varphi_a \circ \Theta$ , the defining property of  $\mathcal{M}$  says that  $F_a$  is a Carleson–Newman Blaschke product, up to the harmless unimodular constant  $-1$ . Direct subtraction gives

$$F_a - \Theta = \frac{-a + \bar{a}\Theta^2}{1 - \bar{a}\Theta}.$$

Because  $|\Theta(z)| \leq 1$  on  $\mathbb{D}$ , the numerator has modulus at most  $2|a|$  and the denominator has modulus at least  $1 - |a|$ . This proves (1).

Fix  $\varepsilon > 0$ . Choose  $a \neq 0$  so small that  $2|a|/(1-|a|) < \varepsilon/2$ . Since  $F_a$  is a finite product of interpolating Blaschke products, the Marshall–Stray theorem provides a single interpolating Blaschke product  $I$  with  $\|I - F_a\|_\infty < \varepsilon/2$ . The triangle inequality now yields

$$\|I - \Theta\|_\infty \leq \|I - F_a\|_\infty + \|F_a - \Theta\|_\infty < \varepsilon.$$

□

**Corollary 1.** *Every Blaschke product whose zeros lie in a Stolz angle is uniformly approximable by interpolating Blaschke products. This includes Blaschke products which are not Carleson–Newman.*

*Proof.* Borichev, Nicolau, Ounaïes, and Thomas record that every Blaschke product whose zeros lie in a Stolz angle belongs to  $\mathcal{M}$  [3, Section 6.5]. Apply the theorem. Strictness is witnessed by their explicit example with zeros  $1 - 2^{-j}$  of multiplicity  $j$ ,  $j \geq 1$ : it belongs to  $\mathcal{M}$  but is not Carleson–Newman [3, Proposition 6.11 and the paragraph following it]. □

## Identification and limitations

The proof is complete, but its inputs are literature theorems and the link is a direct identification. The supporting authors do not explicitly advertise this exact implication as an answer to the question in [1]; hence the provenance is “literature implied,” not “literature already answered.” The result reaches beyond the obvious Carleson–Newman target class, as the Stolz-angle example shows, but it does not establish that an arbitrary Blaschke product lies in  $\mathcal{M}$ . The full density question therefore remains open.

## Search record

The exact question and close variants were searched through arXiv and primary publisher sources using the phrases “every Blaschke product,” “uniform approximation,” “interpolating Blaschke product,” “Frostman shift,” and “Carleson–Newman.” The search located the 2006 modulus theorem of Hjelle and Nicolau, the 2023 open-status statement of Xin and Hou, the Marshall–Stray approximation theorem, and the 2025/2026 SIP– $\mathcal{M}$  results. No full answer was found. The theorem above is the direct combination that survived proof checking.

## References

- [1] I. Chalendar, P. Gorkin, and J. R. Partington, *Inner functions and operator theory*, arXiv:1503.05461 (2015).
- [2] D. E. Marshall and A. Stray, *Interpolating Blaschke products*, Pacific J. Math. **173** (1996), 491–499.
- [3] A. Borichev, A. Nicolau, M. Ounaïes, and P. J. Thomas, *Sharp invertibility in quotient algebras of  $H^\infty$* , arXiv:2501.05143 (2025).
- [4] Y. Xin and B. Hou, *A note on connectedness of Blaschke products*, arXiv:2302.00830 (2023).